

PRODUCT REQUIREMENTS DOCUMENT

Egypt Event Discovery & Ticketing Platform - MVP Features

Document Version: 1.0

Date: February 13, 2026

Project Phase: MVP Development (6 Months)

Target Markets: Egypt (Cairo & Alexandria) → GCC Expansion

Document Owner: Product Team

Classification: Internal Strategic Document

EXECUTIVE SUMMARY

Product Overview

This document defines the Minimum Viable Product (MVP) requirements for Egypt's first integrated event discovery, ticketing, and venue marketplace platform. The platform will connect three distinct user groups—event attendees, event organizers, and venue owners—in a unified ecosystem, addressing the current market fragmentation where ticketing and venue discovery exist as separate services.

Problem Statement

Current Market Challenges:

- Event organizers in Egypt face high commission rates (10-15%) from international ticketing platforms
- No integrated platform connects event discovery with venue booking
- Attendees struggle to discover local events due to fragmented information sources

- Venue owners lack effective channels to reach event organizers
- Complex pricing negotiations delay event planning timelines
- International platforms lack localized payment methods (Fawry, Paymob) and Arabic language support

Solution

A three-sided marketplace platform offering:

- **For Attendees:** Centralized event discovery with seamless ticket purchasing (QR-based mobile tickets)
- **For Organizers:** Low-commission ticketing (3-5% vs. industry 10-15%), integrated marketing tools, and venue discovery
- **For Venue Owners:** Free venue listings with premium promotion options, standardized pricing display, and direct organizer connections

Business Model

Primary Revenue Streams:

1. **Ticket Sales Commission:** 3-5% per ticket (organizer pays)
2. **Venue Premium Listings:** 500 EGP/month for featured placement
3. **Venue Booking Commission:** 5-10% on rental transactions (Phase 2)
4. **Organizer Subscriptions:** Freemium model with Pro tier (Phase 2)

Success Metrics (6-Month MVP Period)

Metric	Month 3 Target	Month 6 Target
Venues Listed	50	100
Active Organizers	10	30
Events Posted	15	50
Tickets Sold	500	2,000
Registered Attendees	300	1,500
Platform GMV (EGP)	75,000	300,000
Platform Revenue (EGP)	3,000	15,000

Table 1: MVP Key Performance Indicators

TARGET AUDIENCE

Primary User Personas

Persona 1: Event Attendee (Demand Side - Priority Focus)

Demographics:

- Age: 18-45 years old
- Location: Cairo, Alexandria
- Tech-savvy, smartphone-first users
- Income: Middle to upper-middle class

Behavioral Characteristics:

- Discovers events through social media (Facebook, Instagram)
- Prefers mobile ticketing over physical tickets
- Values transparent pricing and secure payment methods
- Attends 2-5 events per month (concerts, conferences, cultural events)

Pain Points:

- Difficulty finding comprehensive event listings in Egypt
- Distrust of online payment systems (need local payment gateways)
- Language preference: Arabic-first, but English literate
- Concerns about counterfeit tickets

Goals:

- Easily discover events matching personal interests
- Purchase tickets quickly (<60 seconds)
- Receive instant ticket confirmation via SMS/email
- Smooth entry experience with QR code scanning

Persona 2: Event Organizer (Supply Side)

Demographics:

- Professional event planners, entertainment companies, corporate event managers
- Age: 25-50 years old
- Experience: 1-10 years in event management
- Team size: Solo operators to 10+ person teams

Behavioral Characteristics:

- Organizes 5-20 events per year
- Currently uses international platforms (high costs) or manual ticket sales
- Needs marketing support beyond just ticketing
- Values detailed analytics and attendee data

Pain Points:

- High commission rates (10-15%) erode profit margins
- Difficulty finding suitable venues within budget
- Time-consuming manual attendee tracking and check-in
- Limited marketing reach for locally-focused events

Goals:

- Maximize ticket revenue through lower commission rates
- Access transparent venue pricing and instant booking
- Track sales performance in real-time
- Efficiently manage event check-in process
- Build attendee database for future events

Persona 3: Venue Owner

Demographics:

- Hotels, conference centers, private event spaces, cultural centers
- Business size: Small venues (50-100 capacity) to large (1,000+ capacity)
- Location: Concentrated in Cairo and Alexandria
- Decision-maker: Venue managers, business owners

Behavioral Characteristics:

- Relies on word-of-mouth and personal networks for bookings
- Struggles with visibility to potential clients
- Pricing often negotiated case-by-case (inconsistent)
- Limited digital presence

Pain Points:

- Low online discoverability by event organizers
- Inconsistent booking rates (feast-or-famine cycle)
- Time spent on inquiry responses and price negotiations
- No standardized rental terms or contracts

Goals:

- Increase venue utilization rate
 - Reach new event organizers beyond existing network
 - Establish transparent, market-competitive pricing
 - Reduce administrative burden of booking management
 - Generate consistent revenue stream
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PRODUCT SCOPE

In-Scope for MVP (6 Months)

Core Platform Functions:

- User registration and authentication (three distinct roles)
- Bilingual interface (Arabic/English) with RTL support
- Progressive Web App (PWA) for mobile-first experience
- Integration with Egyptian payment gateways (Paymob, Fawry)
- Email and SMS notification systems
- Admin panel for platform moderation

Attendee-Facing Features:

- Event browsing and search functionality
- Event detail pages with rich media
- Ticket purchasing flow (general admission)
- Promo code redemption

- QR code ticket delivery
- User profile and purchase history
- Social sharing capabilities

Organizer-Facing Features:

- Event creation and management
- Multiple ticket type configuration
- Real-time sales dashboard
- Attendee list export (CSV)
- Basic promotional code generation
- Web-based QR check-in scanner
- Payout tracking interface

Venue Owner Features:

- Venue listing creation (photos, specifications, amenities)
- Standardized pricing display
- Inquiry management system
- Basic venue owner dashboard

Out-of-Scope for MVP (Deferred to Phase 2)

Advanced Ticketing:

- Reserved seating with seat maps
- Dynamic pricing algorithms
- Season passes and multi-event bundles
- Group ticket packages

Advanced Discovery:

- AI-powered event recommendations
- Personalized user feeds
- Interactive map view
- Wishlist and favorites functionality

Advanced Venue Features:

- Instant booking with automated contracts
- Real-time availability calendars
- 3D virtual venue tours

- AI-powered organizer-venue matching

Marketing Automation:

- Email campaign builder
- Social media scheduling
- Attendance prediction analytics
- Affiliate/referral programs

Technology Enhancements:

- Native mobile applications (iOS/Android)
- Offline check-in mode
- Point-of-sale (POS) system for door sales
- Virtual/hybrid event streaming
- Blockchain/NFT ticketing

Geographic Expansion:

- Additional Egyptian cities beyond Cairo/Alexandria
 - GCC market entry (Saudi Arabia, UAE, etc.)
 - Multi-currency support
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FUNCTIONAL REQUIREMENTS

1. USER MANAGEMENT SYSTEM

1.1 User Registration

Priority: P0 (Must Have)

User Story: As a new user, I want to create an account so that I can access platform features relevant to my role.

Acceptance Criteria:

- User can register via email and password
- User must select one of three roles: Attendee, Organizer, or Venue Owner
- Password must meet security requirements (minimum 8 characters, 1 uppercase, 1 number, 1 special character)
- Email verification sent immediately upon registration

- Account remains inactive until email verification completed
- User receives welcome email with onboarding instructions
- Registration form validates data in real-time (duplicate email detection, format validation)
- Phone number collection optional but encouraged

Technical Specifications:

- Backend: Node.js with bcrypt password hashing
- Database: PostgreSQL users table with role enumeration
- Email service: SendGrid API integration
- Validation: Joi schema validation library

1.2 Social Login

Priority: P1 (Should Have)

User Story: As a potential user, I want to sign up using my Facebook or Google account so that I can access the platform faster without creating new credentials.

Acceptance Criteria:

- "Continue with Facebook" and "Continue with Google" buttons displayed on registration/login pages
- OAuth 2.0 authentication flow properly implemented
- User profile data (name, email, profile picture) automatically populated from social account
- User still required to select role (Attendee/Organizer/Venue Owner) after social login
- Account linking: Users can connect social accounts to existing email-based accounts
- Privacy policy and data usage consent displayed before social login completion

1.3 User Authentication

Priority: P0 (Must Have)

User Story: As a registered user, I want to securely log in to my account so that I can access my personalized dashboard and features.

Acceptance Criteria:

- Login form accepts email and password
- Session management with JWT (JSON Web Tokens) - 7-day expiration
- "Remember Me" option extends session to 30 days
- Failed login attempts logged (rate limiting after 5 failed attempts within 15 minutes)
- Clear error messages for invalid credentials without revealing which field is incorrect (security)
- Logout functionality immediately invalidates session token

1.4 Password Reset

Priority: P0 (Must Have)

User Story: As a user who forgot my password, I want to reset it via email so that I can regain access to my account.

Acceptance Criteria:

- "Forgot Password?" link prominently displayed on login page
- User enters email address to receive reset link
- Password reset email sent within 2 minutes
- Reset link valid for 1 hour only
- Reset link is single-use (invalidated after successful password change)
- New password must meet same security requirements as registration
- Confirmation email sent after successful password reset

1.5 Phone Verification (SMS OTP)

Priority: P1 (Should Have)

User Story: As an Egyptian user, I want to verify my account using my mobile number so that I receive ticket confirmations and event updates via SMS.

Acceptance Criteria:

- Phone verification optional during registration but required before first ticket purchase
- User enters Egyptian mobile number (+20 format)

- 6-digit OTP code sent via SMS within 30 seconds
- OTP valid for 5 minutes
- User can request new OTP after 60-second cooldown period
- Maximum 3 OTP requests per phone number per hour (prevent abuse)
- Phone numbers must be unique (one phone per account)

Technical Specifications:

- SMS Gateway: Vodafone Egypt SMS API or Twilio (Egypt-supported)
- OTP storage: Redis with TTL (time-to-live) expiration

1.6 User Profile Management

Priority: P0 (Must Have)

User Story: As a registered user, I want to edit my profile information so that my account details remain current and accurate.

Acceptance Criteria:

- Accessible profile editing page from user dashboard
 - Editable fields: Full name, profile picture, phone number, email (requires re-verification if changed)
 - Role-specific fields:
 - Organizers: Company name, website URL, social media links, business description
 - Venue Owners: Venue name, business license number, contact person
 - Attendees: Date of birth, gender (optional), event preferences
 - Profile picture upload (max 5MB, JPG/PNG formats)
 - Image automatically resized and optimized (Cloudinary integration)
 - Changes saved immediately with success confirmation
 - Profile completion percentage displayed (encourages full completion)
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2. EVENT DISCOVERY & BROWSING

2.1 Event Listing Page

Priority: P0 (Must Have)

User Story: As an attendee, I want to browse all upcoming events so that I can discover interesting activities in my city.

Acceptance Criteria:

- Events displayed in grid view (3 columns desktop, 2 columns tablet, 1 column mobile)
- Each event card shows: Event image, title, date/time, venue name, starting price, category tag
- Events sorted by date (soonest first) by default
- Alternative sorting options: Price (low to high), Popularity (most viewed), Recently Added
- Pagination: 24 events per page with "Load More" button
- Skeleton loading animation during data fetch
- Empty state message when no events match criteria
- Performance: Page loads within 2 seconds on 4G connection

2.2 Search Functionality

Priority: P0 (Must Have)

User Story: As an attendee, I want to search for events by keyword so that I can quickly find specific events or topics of interest.

Acceptance Criteria:

- Search bar prominently placed in header (visible on all pages)
- Search by event title, description, venue name, organizer name, or category
- Real-time search suggestions appear after 3 characters typed (autocomplete)
- Search results update immediately upon query submission
- Highlighted keywords in search results (improved scannability)
- Recent searches saved (last 5 searches) for logged-in users
- Search analytics tracked for popular queries (admin insights)

Technical Specifications:

- Full-text search using PostgreSQL tsvector and tsquery
- Arabic language stemming and tokenization support
- Search index updated via database triggers on event creation/update

2.3 Filter System

Priority: P0 (Must Have)

User Story: As an attendee, I want to filter events by date, location, and category so that I can find events that match my specific preferences.

Acceptance Criteria:

- Filter sidebar visible on left side (desktop) or collapsible panel (mobile)
- **Date Filter:**
 - Predefined options: Today, Tomorrow, This Weekend, This Week, This Month
 - Custom date range picker (start and end date)
 - Calendar view with available event dates highlighted
- **Location Filter:**
 - Dropdown with options: Cairo, Alexandria, Online Events
 - District-level filtering for Cairo (e.g., Zamalek, Maadi, New Cairo)
- **Category Filter:**
 - Multi-select checkboxes
 - Categories: Music & Concerts, Conferences & Business, Sports & Fitness, Arts & Culture, Food & Drink, Workshops & Classes, Nightlife & Parties, Family & Kids, Technology & Innovation, Other
- Active filters displayed as removable tags above event listings
- "Clear All Filters" button resets to default view
- Filter selections persist during session (saved in URL parameters for shareability)
- Results update within 1 second of filter change

2.4 Event Detail Page

Priority: P0 (Must Have)

User Story: As an attendee, I want to view comprehensive event information so that I can decide whether to purchase tickets.

Acceptance Criteria:

- Event detail page accessible via click on event card
- SEO-friendly URL structure: /events/[event-slug]
- **Displayed Information:**
 - Hero image (large, high-quality)
 - Event title (H1 heading)
 - Date and time (with timezone: Cairo Time GMT+2)
 - Venue name with address (clickable for map integration)
 - Organizer name with profile link
 - Category tags
 - Full event description (rich text formatting: bold, italics, bullet points, links)
 - Photo gallery (if multiple images provided)
 - Ticket types with prices and availability counts
 - Event terms and conditions (refund policy, age restrictions, dress code)
 - Social sharing buttons (Facebook, Instagram, WhatsApp, Twitter, Copy Link)
- **Map Integration:**
 - Embedded map showing venue location (Google Maps iframe)
 - "Get Directions" button opens navigation in user's preferred map app
- **Call-to-Action:**
 - "Buy Tickets" button sticky at bottom on mobile
 - Button disabled if event is sold out or past date
 - Shows "Sold Out" or "Event Ended" status clearly
- Related events section at bottom (same category or venue)
- Page view counter visible to organizers only (not public)

Technical Specifications:

- Meta tags optimized for social media sharing (Open Graph, Twitter Cards)
- Structured data markup (JSON-LD) for Google Events rich results
- Image lazy loading for performance

2.5 Social Sharing

Priority: P1 (Should Have)

User Story: As an attendee, I want to share event details on social media so that I can invite friends to attend with me.

Acceptance Criteria:

- Social share buttons for: Facebook, Instagram (story link), WhatsApp, Twitter, LinkedIn, Copy Link
- Clicking share button opens platform-specific sharing interface
- Shared links include: Event image, title, date, venue, and ticket purchase link
- WhatsApp share message pre-populated with event details in Arabic or English based on user language
- "Link Copied" confirmation message after copying link to clipboard
- Share analytics tracked per event (visible to organizers in dashboard)

3. TICKETING ENGINE

3.1 Event Creation (Organizer)

Priority: P0 (Must Have)

User Story: As an event organizer, I want to create a new event listing so that attendees can discover and purchase tickets.

Acceptance Criteria:

- Multi-step event creation form (3 steps: Basic Info, Tickets, Review)
- **Step 1: Basic Information:**
 - Event title (max 100 characters, bilingual: Arabic + English)
 - Event description (rich text editor, max 5,000 characters)

- Event category (dropdown selection)
- Event date and time (date picker + time picker)
- Event duration (optional, in hours)
- Event image upload (required, min 1200x630px, max 10MB)
- Additional images (optional, up to 5 images)
- Venue selection (dropdown of registered venues OR manual address entry for unlisted venues)
- Event capacity (total attendees)
- Age restriction (All Ages, 18+, 21+)
- **Step 2: Ticket Configuration:**
 - Add multiple ticket types (e.g., Early Bird, General Admission, VIP)
 - Each ticket type requires: Name, Price (EGP), Quantity Available, Sales Start Date, Sales End Date
 - Ticket visibility toggle (hide ticket type until sales start)
 - Commission rate display (organizer sees platform fee: 3-5% based on volume tier)
 - Free ticket option (0 EGP, for invitations or promotions)
- **Step 3: Review & Publish:**
 - Preview of complete event listing (as attendees will see it)
 - Edit buttons to return to previous steps
 - Terms acceptance checkbox (organizer agrees to commission structure, refund policy, platform terms)
 - "Publish Event" button submits for admin approval
- Form validation at each step (cannot proceed with incomplete data)
- Draft save functionality (organizers can save progress and return later)
- Event status: Draft → Pending Approval → Published → Past

Technical Specifications:

- Rich text editor: Quill.js or TinyMCE
- Image upload: Cloudinary API with automatic optimization
- Database: Events table with foreign keys to users (organizer), venues, ticket_types (one-to-many)

3.2 Ticket Purchase Flow (Attendee)

Priority: P0 (Must Have)

User Story: As an attendee, I want to purchase event tickets quickly and securely so that I can secure my spot at an event.

Acceptance Criteria:

- **Step 1: Ticket Selection**
 - On event detail page, attendee selects ticket type and quantity
 - Quantity selector (dropdown: 1-10 tickets)
 - Real-time price calculation displayed (subtotal)
 - "Add to Cart" button (cart icon updates with item count)
 - Continue shopping or "Proceed to Checkout" button
- **Step 2: Checkout Page**
 - Cart summary: Event name, ticket type, quantity, unit price, subtotal
 - Promo code entry field (if applicable)
 - Order total displayed prominently (including platform fee if passed to attendee)
 - Attendee information form (if not logged in, required to login or guest checkout)
 - Guest checkout collects: Full name, Email, Phone number
 - Email and phone verification (OTP for phone if SMS verification enabled)
- **Step 3: Payment**
 - Payment method selection: Credit/Debit Card (Paymob), Mobile Wallet (Vodafone Cash, etc.), Fawry (pay at retail locations)
 - Secure payment iframe embedded (Paymob hosted checkout)
 - Payment processing screen with loader (prevent duplicate submissions)
 - Payment timeout after 15 minutes (tickets released back to inventory)
- **Step 4: Confirmation**
 - Order confirmation page with order number
 - QR code tickets displayed (one per ticket purchased)

- Downloadable PDF tickets (all tickets in single file)
- Email confirmation sent immediately with ticket attachments
- SMS confirmation sent with order number and link to tickets
- "Add to Calendar" button (generates .ics file for Google Calendar, Apple Calendar)
- Target completion time: <60 seconds from ticket selection to confirmation
- Abandoned cart recovery: Email sent after 24 hours if user added tickets but didn't complete purchase (Phase 2)

Technical Specifications:

- Inventory locking: Redis to reserve tickets during checkout process
- Payment gateway: Paymob Checkout API integration
- QR code generation: qrcode npm library
- PDF generation: Puppeteer or PDFKit
- Transaction atomicity: Database transactions to ensure payment-ticket consistency

3.3 Ticket Types & Pricing

Priority: P0 (Must Have - General Admission), P1 (Should Have - Multiple Tiers)

User Story: As an organizer, I want to create multiple ticket types with different prices so that I can maximize revenue through tiered pricing strategies.

Acceptance Criteria:

- Organizers can create unlimited ticket types per event
- Each ticket type is independent with own pricing, quantity, and sales period
- **Supported Ticket Types:**
 - General Admission (P0 - MVP required)
 - Early Bird (discounted, limited quantity or time-bound) (P1)
 - VIP/Premium (higher price, additional benefits) (P1)
 - Free Tickets (for promotional purposes) (P1)

- Ticket type display order controlled by organizer (drag-and-drop reordering in dashboard)
- Ticket types can be hidden until specific sale start date/time
- Sold-out ticket types displayed as unavailable (grayed out, cannot select)
- Ticket type description field (optional, up to 500 characters explaining benefits)

Deferred to Phase 2:

- Reserved seating (seat selection from venue map)
- Group bundles (e.g., "Buy 5, Get 1 Free")
- Dynamic pricing (automated price adjustments based on demand)

3.4 QR Code Ticket Generation

Priority: P0 (Must Have)

User Story: As an attendee, I want to receive a QR code ticket on my phone so that I can easily enter the event without printing physical tickets.

Acceptance Criteria:

- Unique QR code generated for each individual ticket (not per order)
- QR code contains: Order ID, Ticket ID, Event ID, Attendee Name, Security Hash
- QR code format: PNG image, 500x500px resolution
- QR code embedded in:
 - Email confirmation (inline image)
 - PDF ticket download
 - Web-based ticket view (accessible via user account "My Tickets" page)
- Ticket design includes: Platform logo, event name, date/time, venue, attendee name, ticket type, QR code, order number, terms
- Bilingual ticket (Arabic and English text)
- Security watermark or background pattern to prevent screenshot fraud (optional Phase 2 enhancement)

Technical Specifications:

- QR encoding: SHA-256 hash of ticket data + secret salt
- Library: qrcode (npm) or node-qrcode
- Storage: QR code images stored in Cloudinary CDN
- Ticket ID format: 12-character alphanumeric (e.g., TKT-A7B3D9E2)

3.5 Promo Code System

Priority: P1 (Should Have)

User Story: As an organizer, I want to create promotional discount codes so that I can incentivize early ticket purchases and reward loyal customers.

Acceptance Criteria:

- Organizers can create promo codes from event dashboard
- **Promo Code Configuration:**
 - Code name (e.g., "EARLYBIRD", "LAUNCH50") - alphanumeric, no spaces, max 20 characters
 - Discount type: Percentage (e.g., 20% off) OR Fixed Amount (e.g., 50 EGP off)
 - Discount value
 - Usage limit: Unlimited OR specific number (e.g., first 100 uses)
 - Validity period: Start date/time and end date/time
 - Applicable ticket types: All tickets OR specific types only
 - Minimum order value (optional, e.g., discount applies only to orders >200 EGP)
- Attendees enter promo code during checkout
- Real-time validation: "Promo code applied!" success message or error (invalid/expired/limit reached)
- Discount displayed in order summary with strikethrough original price
- Promo code usage tracked (organizers see: Total uses, Total discount given, Revenue impact)
- Promo codes case-insensitive
- One promo code per order (cannot stack multiple codes)

Technical Specifications:

- Database: promo_codes table with foreign key to events
 - Validation: Check expiry, usage count, minimum order value before applying discount
 - Transaction recording: Log each promo code use in promo_usage table
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4. ORGANIZER DASHBOARD

4.1 Event Management Interface

Priority: P0 (Must Have)

User Story: As an organizer, I want to view and manage all my events from a single dashboard so that I can efficiently oversee my event portfolio.

Acceptance Criteria:

- Dashboard accessible from organizer profile menu
- "My Events" page lists all events created by organizer
- Event list shows: Event title, Date, Status (Draft/Pending/Published/Past), Ticket sales count, Revenue
- Filter events by status: All, Upcoming, Past, Drafts
- Search within my events by title
- Action buttons for each event:
 - View (see public event page)
 - Edit (modify event details, tickets)
 - Analytics (see detailed stats)
 - Check-in (launch QR scanner)
 - Duplicate (create new event from existing template)
 - Cancel/Delete (with confirmation modal)
- "Create New Event" button prominently placed
- Quick stats overview at top: Total events, Total tickets sold (all-time), Total revenue, Upcoming events count

4.2 Sales Dashboard & Analytics

Priority: P1 (Should Have - Basic Analytics in MVP)

User Story: As an organizer, I want to view real-time ticket sales data so that I can track event performance and make informed marketing

decisions.

Acceptance Criteria:

- Analytics accessible per event from event management interface
- **Sales Metrics Displayed:**
 - Total tickets sold (count and percentage of capacity)
 - Total revenue (gross, before commission deduction)
 - Net revenue (after platform commission)
 - Average ticket price
 - Sales by ticket type (breakdown chart)
- **Visual Charts:**
 - Sales over time (line chart showing daily ticket sales)
 - Revenue over time (cumulative)
 - Ticket type distribution (pie chart or bar chart)
- **Traffic Metrics:**
 - Event page views (total and unique visitors)
 - Conversion rate (views to ticket purchases)
 - Social shares count
- Date range filter (Last 7 days, Last 30 days, All time, Custom range)
- "Export Report" button downloads PDF summary
- Real-time updates (dashboard refreshes every 30 seconds for active events)

Deferred to Phase 2:

- AI-powered attendance predictions
- Competitor benchmarking (compare performance to similar events)
- Detailed demographic insights (attendee age, gender, location breakdowns)
- Heatmap of purchase times (identify peak buying hours)

Technical Specifications:

- Charting library: Chart.js or Recharts (React)
- Data aggregation: Backend API endpoints with optimized queries (indexed database fields)

4.3 Attendee List Management

Priority: P0 (Must Have)

User Story: As an organizer, I want to download a complete list of ticket purchasers so that I can prepare for check-in and communicate with attendees.

Acceptance Criteria:

- "Attendee List" tab in event dashboard
- Table displays: Attendee name, Email, Phone, Ticket type, Quantity purchased, Order date, Check-in status
- Table sortable by any column (click column header)
- Search attendees by name, email, or order ID
- "Export to CSV" button downloads complete attendee list
- CSV includes all fields plus: Order ID, Ticket ID (for multi-ticket orders), QR code URL
- Export respects current filters/search (can export subset of attendees)
- Privacy compliance: Organizers acknowledge data protection responsibilities before first export

Technical Specifications:

- CSV generation: json2csv npm library
- Data sanitization: Remove sensitive platform data (payment details) from exports

4.4 Promo Code Management

Priority: P1 (Should Have)

User Story: As an organizer, I want to track the performance of my promotional codes so that I can evaluate which marketing strategies are most effective.

Acceptance Criteria:

- "Promo Codes" tab in event dashboard
- List all active and expired promo codes for event

- Displayed metrics per code: Uses count, Total discount given, Revenue generated with code, Remaining uses (if limited)
- "Create New Promo Code" button opens creation modal
- Edit/delete existing promo codes (cannot edit code name, only parameters)
- Disable code button (manually deactivate before expiry date)
- Filter codes: Active, Expired, All
- Export promo code usage report (CSV with order-level details)

4.5 Payout Tracking

Priority: P0 (Must Have)

User Story: As an organizer, I want to see my earnings and payout schedule so that I understand when I will receive event revenue.

Acceptance Criteria:

- "Payouts" section in organizer dashboard
- **Payout Status Categories:**
 - Pending: Revenue from recent ticket sales (within hold period)
 - Available: Ready for payout (hold period completed)
 - Paid: Successfully transferred to organizer's bank account
- Payout schedule clearly stated: "Revenue released 7 days after event end date"
- Breakdown by event: Event name, Gross revenue, Platform commission, Net payout, Status
- Bank account management (organizers add/verify bank details for payouts)
- Payout history: All past transfers with dates and transaction IDs
- "Request Payout" button for available funds (triggers manual admin processing in MVP, automated in Phase 2)
- Expected payout date displayed for pending amounts

Technical Specifications:

- Payout hold period: Configurable (default 7 days post-event for refund buffer)
- Bank account storage: Encrypted database field

- Payment processing: Manual bank transfer in MVP, integrate with payment gateway disbursement API in Phase 2
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5. VENUE MARKETPLACE

5.1 Venue Listing Creation (Venue Owner)

Priority: P1 (Should Have)

User Story: As a venue owner, I want to create a detailed listing for my venue so that event organizers can discover and inquire about renting my space.

Acceptance Criteria:

- Venue listing creation form accessible from venue owner dashboard
- **Required Information:**
 - Venue name (Arabic + English)
 - Venue type: Conference Hall, Hotel, Outdoor Space, Cultural Center, Restaurant/Cafe, Theater, Sports Facility, Other
 - Complete address (street, district, city, governorate)
 - GPS coordinates (auto-populated from address OR manual pin placement on map)
 - Capacity: Standing, Seated, Mixed (specify numbers)
 - Contact person name and phone number
 - Venue description (up to 2,000 characters)
 - Available amenities (multi-select): WiFi, Parking, Catering, Audio/Visual Equipment, Stage, Air Conditioning, Accessible for Disabilities, Security, Kitchen, Outdoor Area
- **Pricing Configuration:**
 - Hourly rate (EGP)
 - Half-day rate (4-6 hours)
 - Full-day rate (8-12 hours)
 - Weekend vs. weekday pricing (optional differential)
 - Minimum booking duration
 - Additional fees (security deposit, cleaning fee, equipment rental)
- **Media Upload:**
 - Minimum 5 high-quality photos (required)

- Maximum 20 photos
- Photo upload order determines display order (drag to reorder)
- Recommended resolution: 1920x1080px
- Accepted formats: JPG, PNG (max 10MB per image)
- **Availability Settings:**
 - Operating hours (e.g., 9 AM - 11 PM)
 - Days of operation (e.g., closed Fridays)
 - Blackout dates (specific dates when venue is unavailable)
- Form validation at submission
- Listing status: Draft → Pending Approval → Published
- Admin approval required before listing goes live (quality control)

5.2 Venue Discovery (Organizer & Attendee)

Priority: P1 (Should Have)

User Story: As an event organizer, I want to browse available venues with transparent pricing so that I can find the perfect location for my event without time-consuming negotiations.

Acceptance Criteria:

- "Venues" navigation link in main menu
- Venue browsing page displays all published venue listings
- Grid view: Venue cards showing primary image, name, location, capacity range, starting price (hourly rate)
- "Featured Venues" section at top (premium listings)
- **Filter Options:**
 - Location: Cairo (districts), Alexandria, Other
 - Venue type (checkboxes)
 - Capacity range (slider: 0-2000)
 - Price range (slider: 0-50,000 EGP per day)
 - Amenities (multi-select checkboxes)
 - Availability date (calendar picker shows only available venues)
- Sorting: Price (low to high), Capacity (small to large), Recently Added, Most Popular
- Search by venue name or location

- Results update in real-time as filters applied

5.3 Venue Detail Page

Priority: P1 (Should Have)

User Story: As an organizer, I want to view complete venue details and pricing so that I can evaluate if the space meets my event requirements.

Acceptance Criteria:

- Venue detail page URL: /venues/[venue-slug]
- **Page Sections:**
 - Photo gallery (large hero image with thumbnail carousel)
 - Venue name and type
 - Location with embedded map
 - Capacity details (standing, seated)
 - Full description
 - Amenities list with icons
 - **Standardized Pricing Table:**

Duration	Price (EGP)
Hourly Rate	[Amount]
Half-Day (4-6 hours)	[Amount]
Full-Day (8-12 hours)	[Amount]
Additional Hour	[Amount]

- Additional fees breakdown (deposit, cleaning, etc.)
 - Availability calendar (shows booked dates grayed out - Phase 2 feature)
 - House rules and policies (cancellation policy, deposit terms)
 - Contact information (phone, email)
- "Inquire About This Venue" button (primary CTA)
- Similar venues section (same location or capacity)
- Breadcrumb navigation: Home > Venues > [Venue Name]

5.4 Venue Inquiry System

Priority: P1 (Should Have - MVP Basic Inquiry Form)

User Story: As an organizer, I want to send booking inquiries to venue owners so that I can check availability and finalize rental details.

Acceptance Criteria:

- Inquiry form modal opens when "Inquire About This Venue" clicked
- **Form Fields:**
 - Organizer name and organization (auto-filled if logged in)
 - Email and phone (auto-filled)
 - Event date (date picker)
 - Event duration (dropdown: 2 hours, 4 hours, 6 hours, Full day, Multiple days)
 - Expected attendance (number input)
 - Event type (dropdown: Conference, Concert, Wedding, Corporate Event, Other)
 - Additional requirements (text area: catering needs, equipment, setup preferences)
 - Budget range (optional)
- Form submission creates inquiry record in database
- Venue owner receives immediate email notification with inquiry details
- Inquiry also appears in venue owner's dashboard "Inquiries" tab
- Organizer receives confirmation email: "Your inquiry has been sent. [Venue Name] will respond within 24-48 hours."
- Inquiry status: Pending, Responded, Booked, Declined
- Venue owners can respond via:
 - In-platform messaging (MVP: Email notification with response)
 - Phone/email direct contact (contact info shared after inquiry submission)

Deferred to Phase 2:

- Instant booking (automated contract generation and payment processing)
- Real-time availability calendar (prevent double-bookings)
- In-platform messaging system (chat interface between organizer and venue owner)
- Automated price quotes based on inquiry parameters

5.5 Venue Owner Dashboard

Priority: P1 (Should Have)

User Story: As a venue owner, I want to manage my venue listings and inquiries from a centralized dashboard so that I can efficiently handle booking requests.

Acceptance Criteria:

- Dashboard sections:
 - **My Venues:** List all venue listings with edit/delete options
 - **Inquiries:** Table of all booking inquiries (pending, responded, booked, declined)
 - **Analytics:** Basic metrics (venue page views, inquiries received, conversion rate)
 - **Premium Subscription:** Option to upgrade to featured listing (500 EGP/month)
- **Inquiry Management:**
 - Filter inquiries by status and date
 - View inquiry details (organizer info, event details, requirements)
 - Action buttons: Mark as Responded, Mark as Booked, Decline
 - Notes field (internal notes not visible to organizer)
- Email notifications for new inquiries (real-time)
- Mobile-responsive dashboard (manage inquiries on-the-go)

5.6 Premium Venue Listings

Priority: P1 (Should Have - Revenue Stream)

User Story: As a venue owner, I want to upgrade my listing to featured status so that my venue appears more prominently and receives more inquiries.

Acceptance Criteria:

- "Upgrade to Premium" option in venue dashboard
 - **Premium Benefits:**
 - Featured placement in "Top Venues" section on homepage
 - Highlighted border and "Featured" badge on venue cards
 - Priority ranking in search results
 - Verified venue checkmark icon
 - Enhanced analytics (detailed traffic sources, user behavior)
 - Pricing: 500 EGP per month per venue
 - Payment via platform (same payment gateways: Paymob, Fawry)
 - Subscription auto-renews monthly unless canceled
 - Cancel anytime (prorated refund not available)
 - Premium status active immediately upon payment confirmation
 - Invoice generation for business accounting
-

6. CHECK-IN SYSTEM

6.1 QR Code Scanner (Web-Based)

Priority: P1 (Should Have - Launch in Month 5-6)

User Story: As an event organizer or staff member, I want to scan attendee QR code tickets so that I can verify entry and prevent duplicate entries.

Acceptance Criteria:

- Web-based QR scanner accessible from organizer event dashboard
- URL format: /check-in/[event-id]
- Device camera access permission requested
- Real-time QR code scanning using device camera (rear or front camera toggle)
- **Scan Result Scenarios:**
 - **Valid Ticket:** Green success screen, attendee name displayed, ticket marked as checked-in, entry granted
 - **Already Used:** Red warning screen, "This ticket was already scanned at [time]", deny entry

- **Invalid QR Code:** Red error screen, "Invalid ticket or wrong event", deny entry
- **Wrong Event:** Red error screen, "This ticket is for [Event Name], not this event"
- Audio feedback: Success beep (valid), error buzz (invalid/duplicate)
- Visual feedback: Color-coded screens (green/red), large text visible from distance
- Attendee counter: "X of Y attendees checked in" displayed on scanner screen
- Check-in log: Timestamped list of all scans (visible in organizer dashboard)
- Offline mode (Phase 2): Scanner works without internet, syncs when connection restored
- Multiple scanner support: Several staff members can scan simultaneously for large events

Technical Specifications:

- QR scanning library: `html5-qrcode` (JavaScript, works in mobile browsers)
- Real-time validation: API call to verify ticket on each scan
- Database update: Mark ticket `checked_in` timestamp immediately
- Race condition handling: Database-level locking to prevent simultaneous duplicate scans

6.2 Manual Check-In (Fallback)

Priority: P2 (Nice to Have)

User Story: As an organizer, I want to manually check in attendees by searching their name or order ID so that I can accommodate attendees who forgot their tickets or have device issues.

Acceptance Criteria:

- "Manual Check-In" tab in check-in interface
- Search by: Attendee name, email, order ID, or ticket ID
- Search results display matching attendees with ticket status
- "Check In" button next to each attendee

- Confirmation prompt before manual check-in (prevent accidental clicks)
 - Manual check-in logged with timestamp and staff member who processed it
 - Export manual check-in log (audit trail)
-

7. PAYMENT INTEGRATION

7.1 Paymob Integration (Primary Gateway)

Priority: P0 (Must Have)

User Story: As an attendee, I want to pay for tickets using my credit/debit card or mobile wallet so that I can complete my purchase conveniently.

Acceptance Criteria:

- Paymob Checkout integration (hosted payment page)
- **Supported Payment Methods:**
 - Visa and Mastercard (credit/debit)
 - Mobile wallets: Vodafone Cash, Orange Cash, Etisalat Cash, Fawry (digital)
 - Installment plans (if Paymob supports for ticket amounts >500 EGP)
- Payment flow:
 1. User clicks "Proceed to Payment" on checkout page
 2. Redirect to Paymob secure payment page (HTTPS)
 3. User completes payment
 4. Paymob redirects back to platform with transaction result
 5. Platform validates transaction via Paymob API callback
 6. Confirmation page displayed (success or failure)
- Transaction fees: 2.75% + 3 EGP per transaction (absorbed by platform, deducted from commission)
- Payment timeout: 15 minutes (ticket inventory released if payment not completed)
- Failed payment handling: Clear error message, option to retry or change payment method
- Payment receipt: Digital receipt sent via email (includes transaction ID, breakdown, payment method)

Technical Specifications:

- Paymob API: RESTful integration with authentication token
- Webhook: Paymob callback URL to receive transaction status updates
- Security: Verify transaction signature (HMAC) to prevent tampering
- Testing: Sandbox environment for development, live credentials for production

7.2 Fawry Integration (Secondary Gateway)

Priority: P1 (Should Have - Add in Month 4-5)

User Story: As an attendee who prefers cash or doesn't have a credit card, I want to pay using Fawry so that I can purchase tickets at nearby retail locations.

Acceptance Criteria:

- Fawry payment option displayed during checkout
- User selects "Pay with Fawry"
- Platform generates Fawry payment reference number (14-digit code)
- Instructions displayed: "Visit any Fawry location and provide this reference number to complete payment"
- Ticket reservation held for 24 hours pending payment
- SMS and email sent with Fawry reference number and nearest locations
- Fawry callback notifies platform when payment completed
- Tickets automatically issued upon payment confirmation
- Reminder SMS sent after 12 hours if payment still pending
- Reservation expires after 24 hours if unpaid (tickets released to inventory)

Technical Specifications:

- Fawry API: Payment reference generation and status polling
- Webhook: Fawry callback for payment confirmation
- Setup fee: 999 EGP one-time merchant registration

7.3 Refund Processing

Priority: P1 (Should Have)

User Story: As an attendee, I want to request a refund if an event is canceled so that I can recover my money.

Acceptance Criteria:

- **Refund Policy:**
 - Event canceled by organizer: Full refund (100%)
 - Event postponed: Option to transfer ticket to new date OR refund (minus 10% processing fee)
 - Attendee cancellation: Per organizer's refund policy (displayed at purchase time)
- Refund request flow:
 1. Attendee clicks "Request Refund" on ticket in "My Tickets" page
 2. Selects reason (dropdown: Event canceled, Cannot attend, Other)
 3. Submits request
 4. Admin reviews request (manual approval in MVP)
 5. Refund processed to original payment method
 6. Confirmation email sent when refund initiated
- Refund processing time: 5-10 business days (as per payment gateway policy)
- Refund status visible in user account: Pending, Approved, Completed, Denied
- Organizer notified of refund requests (impacts their payout)

Technical Specifications:

- Paymob refund API: Initiate refunds via API call
 - Refund record: Database table tracking all refund requests and statuses
 - Accounting: Adjust organizer payouts to account for refunded tickets
-

8. NOTIFICATIONS SYSTEM

8.1 Email Notifications

Priority: P0 (Must Have)

User Story: As a user, I want to receive email notifications for important account and transaction events so that I stay informed.

Acceptance Criteria:

- **Email Triggers:**
 - **Attendees:**
 - * Welcome email (post-registration)
 - * Email verification link
 - * Ticket purchase confirmation (with ticket attachments)
 - * Event reminder (24 hours before event)
 - * Event cancellation notification
 - * Refund status updates
 - **Organizers:**
 - * Event published confirmation
 - * New ticket sale notification (real-time or daily digest option)
 - * Milestone alerts (50%, 75%, 100% sold out)
 - * Event approaching reminder (7 days, 3 days, 1 day before)
 - * Payout processed notification
 - **Venue Owners:**
 - * New inquiry notification (real-time)
 - * Inquiry status change (when organizer follows up)
 - * Premium subscription renewal reminder
- Email template design: Branded (platform logo, colors), bilingual (Arabic/English based on user preference), mobile-responsive
- Unsubscribe link in footer (except transactional emails like ticket confirmations)
- Email deliverability: SPF, DKIM, DMARC records configured
- Open and click tracking (for marketing emails, not transactional)

Technical Specifications:

- Email service: SendGrid (free tier: 100 emails/day) or Mailgun
- Templates: HTML email templates with inline CSS
- Queue: Redis-backed email queue (bull or bee-queue) for asynchronous sending

8.2 SMS Notifications

Priority: P1 (Should Have)

User Story: As an Egyptian user who relies on SMS, I want to receive ticket confirmations and event reminders via SMS so that I have immediate access to my ticket information.

Acceptance Criteria:

- **SMS Triggers:**
 - OTP for phone verification (during registration/checkout)
 - Ticket purchase confirmation with order number and ticket link
 - Event reminder (3 hours before event start time)
 - Urgent notifications (event cancellation, venue change)
- SMS character limit: 160 characters (Arabic: ~70 characters) - concise messages
- Link shortening: Use URL shortener for ticket links (e.g., tkt.eg/ABC123)
- SMS template in Arabic (primary) and English (if user's preferred language is English)
- Opt-out option: Users can disable non-critical SMS (reminders) but not transactional (tickets)
- SMS delivery tracking: Log delivery status (sent, delivered, failed)

Technical Specifications:

- SMS Gateway: Vodafone Egypt SMS API or Twilio (with Egypt phone number support)
- Cost: ~0.10-0.20 EGP per SMS (budget accordingly)
- Rate limiting: Prevent SMS spam, max 5 SMS per user per hour

8.3 Push Notifications (Phase 2 - Native App)

Priority: P2 (Deferred - Requires Native Mobile App)

Description: Browser push notifications for PWA in Phase 2, full push notifications in native mobile apps.

9. PLATFORM ADMINISTRATION

9.1 Admin Dashboard

Priority: P0 (Must Have)

User Story: As a platform administrator, I want a comprehensive dashboard to monitor platform activity and moderate content so that I can maintain quality and security.

Acceptance Criteria:

- Admin login separate from user login (different URL: /admin/login)
- Role-based access control (Super Admin, Moderator roles)
- **Dashboard Overview:**
 - Total users (Attendees, Organizers, Venue Owners) - today, this week, all-time
 - Total events (Pending approval, Published, Past)
 - Total tickets sold - today, this week, this month, all-time
 - Total revenue - GMV (Gross Merchandise Value) and platform commission
 - Active inquiries (venue inquiries pending response)
 - User growth chart (new registrations over time)
 - Sales chart (tickets sold and revenue over time)
- **User Management:**
 - Search users by name, email, role
 - View user details and activity history
 - Ban/suspend users (with reason logging)
 - Delete accounts (with confirmation and data retention policy compliance)
- **Event Moderation:**
 - Queue of pending events (new submissions awaiting approval)

- Review event details (title, description, images, ticket pricing)
- Approve or reject events (rejection requires reason sent to organizer)
- Edit event details if needed (typos, formatting)
- Flag inappropriate content (spam, misleading info)
- **Venue Moderation:**
 - Queue of pending venue listings
 - Review venue details and photos
 - Approve or reject listings
 - Edit venue information if needed
- **Transaction Monitoring:**
 - View all transactions (ticket purchases, refunds, payouts)
 - Filter by date range, status, payment method
 - Search by order ID or user email
 - Export transaction reports (CSV)
 - Flag suspicious transactions for manual review
- **Financial Management:**
 - Revenue reports (platform commission breakdown)
 - Payout queue (organizers awaiting payouts)
 - Process payouts (manual bank transfer in MVP, mark as paid)
 - Refund request queue (approve/deny attendee refund requests)
- **System Settings:**
 - Configure commission rates (3-5% tiers based on ticket volume)
 - Set payout hold period (default 7 days)
 - Manage featured content (homepage banners, featured events/venues)
 - Platform-wide announcements (banner at top of site)

Technical Specifications:

- Admin authentication: Separate JWT tokens with role claim
- Audit logging: All admin actions logged (who, what, when) for accountability
- Database queries optimized with indexes for dashboard performance

9.2 Content Moderation Guidelines

Priority: P0 (Must Have - Documented Policies)

Description: Establish clear guidelines for what constitutes acceptable events, venues, and user behavior.

Event Approval Criteria:

- Event must have clear title and accurate description
- Event images must be high-quality and relevant (no misleading graphics)
- Ticket pricing must be reasonable and transparent (no hidden fees)
- Event must comply with Egyptian laws (no prohibited activities: gambling, alcohol sales without license)
- Contact information must be valid (organizer reachable)
- No duplicate events (same event posted multiple times)
- No spam or promotional content unrelated to event (e.g., affiliate links)

Venue Approval Criteria:

- Venue must be a legitimate, physical location
- Photos must accurately represent the venue (no stock photos from other locations)
- Pricing must be clear and complete
- Contact information must be valid
- No fraudulent listings (verified via business registration or site visit for high-value venues)

User Behavior Policies:

- No harassment or abusive language (in inquiries, reviews, messages)
- No fraudulent transactions or chargebacks abuse
- No impersonation of organizers, venues, or platform staff
- No spamming (mass event creation, excessive inquiries)

Enforcement Actions:

- Warning (first offense, minor violation)

- Temporary suspension (repeated violations, 7-30 days)
 - Permanent ban (serious violations: fraud, illegal content)
 - Content removal (delete violating events/venues)
-

10. LOCALIZATION & INTERNATIONALIZATION

10.1 Bilingual Interface (Arabic & English)

Priority: P0 (Must Have)

User Story: As an Egyptian user, I want to use the platform in Arabic so that I can comfortably browse and purchase tickets in my native language.

Acceptance Criteria:

- Language toggle in header (globe icon, dropdown: العربية / English)
- User's language preference saved (cookie or user account setting)
- All UI elements translated: Navigation, buttons, labels, error messages, help text
- **Translation Coverage:**
 - Homepage
 - Event browsing and detail pages
 - Ticket checkout flow
 - User account pages
 - Organizer dashboard
 - Venue listings
 - Admin panel (English only acceptable for MVP)
- Dynamic content (event titles, descriptions) stored in both languages when organizers provide bilingual input
- Fallback: If Arabic translation missing for user-generated content, display English version with language indicator
- Date and time formatting localized (Arabic numerals in Arabic UI, Western numerals in English UI)
- Currency display: Always EGP (Egyptian Pound) regardless of language

Technical Specifications:

- i18n library: react-i18next (React) or next-i18next (Next.js)
- Translation files: JSON format, organized by page/component
- Translation management: Use spreadsheet or translation management tool (Lokalise, Phrase) for non-technical team members to contribute translations

10.2 RTL (Right-to-Left) Layout Support

Priority: P0 (Must Have)

User Story: As an Arabic reader, I want the interface to flow from right to left so that the layout feels natural and intuitive.

Acceptance Criteria:

- Entire layout mirrors when Arabic language selected
- Text alignment: Right-aligned for Arabic, left-aligned for English
- Navigation menus: Flow from right to left (hamburger menu on left becomes right)
- Forms: Labels and inputs right-aligned, validation messages right-aligned
- Icons and arrows: Flipped (back arrow points right, forward arrow points left)
- Carousels and sliders: Reverse direction (next moves left, previous moves right)
- Flexbox and Grid layouts: Reverse order using CSS direction: rtl
- Tables: Column order reversed (rightmost column is first)
- No UI breakage: All elements remain properly aligned and functional in RTL mode
- Images and logos: Do NOT flip (maintain original orientation)

Technical Specifications:

- CSS approach: Use [dir="rtl"] selector for RTL-specific styles
- Framework support: Tailwind CSS has built-in RTL support (rtl: modifier)
- Testing: Manual testing in Arabic mode on all major pages

10.3 Cultural Localization

Priority: P1 (Should Have)

User Story: As an Egyptian user, I want the platform to respect local cultural norms and calendar so that it feels tailored to my context.

Acceptance Criteria:

- **Calendar Integration:**
 - Hijri calendar option (Islamic calendar) in addition to Gregorian
 - Display both calendars on event detail pages (e.g., "March 15, 2026 / Ramadan 16, 1447")
 - Highlight Islamic holidays and Ramadan dates
- **Weekend Awareness:**
 - Default weekend in Egypt: Friday-Saturday (not Saturday-Sunday)
 - Venue availability and event scheduling defaults respect this
- **Prayer Time Consideration:**
 - Event scheduling recommendations avoid Friday prayer time (12-2 PM)
 - Venue listings note if prayer room available
- **Gender-Specific Events:**
 - Event creation allows organizers to specify gender restrictions (All, Men Only, Women Only, Families Only)
 - Displayed prominently on event page
- **Payment Methods:**
 - Prioritize local payment methods (Fawry, Egyptian mobile wallets) over international (credit cards)

NON-FUNCTIONAL REQUIREMENTS

Performance

Page Load Speed:

- Homepage: <2 seconds (first contentful paint)
- Event listing page: <2 seconds
- Event detail page: <2.5 seconds

- Checkout page: <1.5 seconds (critical conversion point)
- Dashboard pages: <3 seconds
- Measured on 4G mobile connection (average Egypt internet speed)

Technical Targets:

- Lighthouse Performance Score: >85
- Time to Interactive (TTI): <3.5 seconds
- First Input Delay (FID): <100ms
- Cumulative Layout Shift (CLS): <0.1

Optimization Strategies:

- Image optimization: WebP format, lazy loading, responsive images, CDN delivery (Cloudinary)
- Code splitting: Load only JavaScript needed for current page
- Caching: Browser caching (1 year for static assets), Redis caching for API responses
- Database indexing: Index all frequently queried fields (user_id, event_id, date ranges)
- Minification: CSS, JavaScript, HTML minified in production
- Compression: Gzip or Brotli compression for text assets

Scalability

User Capacity:

- MVP target: Support 10,000 registered users, 5,000 monthly active users
- Phase 2: Scale to 100,000 registered users, 50,000 monthly active users

Concurrent Users:

- MVP: 500 concurrent users without performance degradation
- Phase 2: 5,000 concurrent users

Transaction Volume:

- MVP: 100 ticket transactions per hour during peak times (event launch, sales start)

- Phase 2: 1,000 transactions per hour

Infrastructure:

- Horizontal scaling: Stateless application servers (can add more servers as needed)
- Database: PostgreSQL with read replicas for scaling read-heavy operations
- Load balancing: NGINX or Cloudflare load balancer
- CDN: Cloudflare for static asset delivery (global edge network)

Security

Authentication & Authorization:

- Password hashing: bcrypt with salt (work factor: 12)
- JWT tokens: Short expiration (7 days), secure HTTP-only cookies
- Role-based access control (RBAC): Strict permission checks on all API endpoints
- Session management: Single sign-on (SSO) support for Phase 2
- Failed login protection: Rate limiting (5 attempts per 15 minutes), account lockout after 10 failed attempts

Data Protection:

- HTTPS/TLS: All traffic encrypted (SSL certificate)
- Database encryption: Encrypt sensitive fields (bank account numbers, phone numbers) at rest
- PCI DSS Compliance: Use Paymob/Fawry hosted payment pages (platform never stores card numbers)
- GDPR/Data Privacy: User data export and deletion on request (right to be forgotten)
- Backup: Daily database backups with 30-day retention, encrypted backup storage

Fraud Prevention:

- QR code security: SHA-256 signed QR codes with secret salt
- Duplicate entry prevention: Database-level constraints (unique ticket check-ins)
- Rate limiting: API rate limits (100 requests per minute per IP), prevent DDoS and scraping

- Transaction monitoring: Flag unusual patterns (bulk ticket purchases from single IP, rapid successive transactions)
- Chargeback protection: 7-day payout hold period, clear refund policies

Security Testing:

- SQL injection prevention: Parameterized queries, ORM usage (Sequelize, TypeORM)
- XSS prevention: Input sanitization, output encoding, Content Security Policy (CSP) headers
- CSRF protection: CSRF tokens on all state-changing requests
- Dependency scanning: Regular updates, vulnerability scanning (npm audit, Snyk)
- Penetration testing: Third-party security audit before public launch

Reliability & Availability

Uptime Target:

- MVP: 99% uptime (allows ~7 hours downtime per month)
- Phase 2: 99.9% uptime (allows ~43 minutes downtime per month)

Backup & Recovery:

- Database backups: Daily full backup, hourly incremental backups
- Recovery Time Objective (RTO): <4 hours (time to restore service after failure)
- Recovery Point Objective (RPO): <1 hour (maximum data loss acceptable)
- Disaster recovery plan: Off-site backup storage, documented restoration procedures

Error Handling:

- Graceful degradation: Platform remains functional even if non-critical services fail (e.g., SMS service down, email notifications still work)

- User-friendly error messages: Clear, actionable error messages (avoid technical jargon)
- Error logging: Centralized logging (Sentry, LogRocket) for monitoring and debugging
- Alerting: Real-time alerts to development team for critical errors (payment failures, database errors)

Usability & Accessibility

Usability Standards:

- Mobile-first design: 70%+ of Egypt users access via mobile
- Touch-friendly: Buttons and interactive elements minimum 44x44px tap targets
- Intuitive navigation: Maximum 3 clicks to reach any page from homepage
- Consistent UI: Design system with reusable components, consistent spacing and typography
- Feedback: Immediate visual feedback for all user actions (loading states, success/error messages)

Accessibility (WCAG 2.1 AA Compliance):

- Keyboard navigation: All interactive elements accessible via keyboard (Tab, Enter, Arrow keys)
- Screen reader support: Semantic HTML, ARIA labels where needed, descriptive alt text for images
- Color contrast: Minimum 4.5:1 for normal text, 3:1 for large text (18px+ or 14px bold)
- Focus indicators: Visible focus state for keyboard users (blue outline or similar)
- Form labels: All inputs have associated labels, error messages linked to fields
- Skip links: "Skip to main content" link for screen reader users
- Resize support: Text remains readable at 200% zoom

Browser & Device Compatibility

Supported Browsers:

- Chrome: Latest 2 versions (primary browser in Egypt)
- Firefox: Latest 2 versions
- Safari: Latest 2 versions (iOS users)
- Edge: Latest 2 versions
- Opera: Latest version
- UC Browser: Latest version (popular in Egypt)

Supported Devices:

- Smartphones: iOS 13+, Android 8+
- Tablets: iPad, Android tablets
- Desktop: Windows 10+, macOS 10.14+
- Screen sizes: 320px (iPhone SE) to 2560px (large desktop)

Progressive Web App (PWA) Requirements:

- Installable: Users can add to home screen (iOS and Android)
- App manifest: Configured with app name, icons, theme colors
- Service worker: Cache static assets for faster load times (full offline support in Phase 2)
- HTTPS: Required for PWA features
- Responsive: Single codebase works across all devices

SEO & Discoverability

Search Engine Optimization:

- Server-side rendering (SSR): Next.js for SEO-friendly page generation
- Meta tags: Unique title, description, Open Graph tags for each page
- Structured data: [Schema.org](https://schema.org) JSON-LD markup for events (Google Events rich results)
- Sitemap: XML sitemap submitted to Google Search Console
- Robots.txt: Configured to allow search engine crawling
- URL structure: Clean, descriptive URLs (/events/concert-name, not /event?id=123)

- Page speed: Fast loading times (Google ranking factor)
- Mobile-friendly: Responsive design (Google mobile-first indexing)
- Canonical URLs: Prevent duplicate content issues

Content Strategy:

- Event categories: Clear categorization for better organization and filtering
 - Keyword optimization: Event titles and descriptions encourage descriptive language
 - Fresh content: Regularly updated with new events (signals active site to Google)
 - Internal linking: Related events, category pages, venue pages interlinked
-

TECHNICAL ARCHITECTURE

Technology Stack

Frontend:

- **Framework:** Next.js 14 (React framework with SSR, SSG, and API routes)
- **UI Library:** React 18 with TypeScript
- **Styling:** Tailwind CSS (utility-first, rapid development, built-in RTL support)
- **State Management:** React Context API + useReducer (simpler than Redux for MVP)
- **Forms:** React Hook Form (performant, easy validation)
- **HTTP Client:** Axios (API requests with interceptors)
- **Date Handling:** date-fns (lightweight, tree-shakeable)
- **Internationalization:** next-i18next (SSR-compatible i18n)
- **Charts:** Chart.js or Recharts (analytics dashboards)

Backend:

- **Runtime:** Node.js 20 LTS
- **Framework:** Express.js (lightweight, flexible)
- **Language:** TypeScript (type safety, better developer experience)

- **API Design:** RESTful API (Phase 2 can add GraphQL for complex queries)
- **Authentication:** JSON Web Tokens (JWT) with jsonwebtoken library
- **Validation:** Joi or Zod (schema validation)
- **ORM:** Prisma or TypeORM (type-safe database access)

Database:

- **Primary Database:** PostgreSQL 15 (relational data: users, events, tickets, transactions)
- **Caching:** Redis 7 (session storage, ticket inventory locking, rate limiting)
- **Search:** PostgreSQL full-text search (upgrade to Elasticsearch in Phase 2 if needed)

File Storage:

- **Media Storage:** Cloudinary (image uploads, automatic optimization, CDN delivery)
- **Document Storage:** AWS S3 or DigitalOcean Spaces (ticket PDFs, exports)

Payment Gateways:

- Paymob (primary)
- Fawry (secondary)

Communication:

- **Email:** SendGrid or Mailgun
- **SMS:** Vodafone Egypt SMS API or Twilio

Infrastructure:

- **Hosting:** DigitalOcean Droplets (cost-effective) OR Railway (managed platform)
- **CDN:** Cloudflare (free tier, DDoS protection, SSL)
- **Domain:** .eg domain (Egyptian TLD for local credibility) or .com
- **SSL Certificate:** Let's Encrypt (free) or Cloudflare SSL
- **Monitoring:** Sentry (error tracking), Google Analytics (usage analytics)

Development Tools:

- **Version Control:** Git + GitHub/GitLab
- **CI/CD:** GitHub Actions or GitLab CI (automated testing and deployment)
- **Code Quality:** ESLint, Prettier (linting and formatting)
- **Testing:** Jest (unit tests), Cypress (end-to-end tests) - Phase 2 priority

Database Schema (High-Level)

Core Tables:

- **users:** id, email, password_hash, role (enum: attendee, organizer, venue_owner), name, phone, email_verified, phone_verified, created_at
- **events:** id, organizer_id (FK users), title_en, title_ar, description_en, description_ar, category, venue_id (FK venues), event_date, capacity, status (enum: draft, pending, published, past, canceled), created_at
- **ticket_types:** id, event_id (FK events), name, price, quantity_total, quantity_sold, sales_start, sales_end, created_at
- **orders:** id, attendee_id (FK users), event_id, total_amount, commission, status (enum: pending, completed, refunded), payment_method, created_at
- **tickets:** id, order_id (FK orders), ticket_type_id (FK ticket_types), qr_code_url, attendee_name, checked_in_at, created_at
- **venues:** id, owner_id (FK users), name_en, name_ar, type, address, city, capacity_standing, capacity_seated, hourly_rate, daily_rate, amenities (JSON), status (enum: pending, published), created_at
- **venue_inquiries:** id, venue_id (FK venues), organizer_id (FK users), event_date, duration, attendance, event_type, message, status (enum: pending, responded, booked, declined), created_at
- **promo_codes:** id, event_id (FK events), code, discount_type (enum: percentage, fixed), discount_value, usage_limit, usage_count, valid_from, valid_until, created_at
- **transactions:** id, order_id (FK orders), amount, payment_gateway, gateway_transaction_id, status (enum: pending, success, failed, refunded), created_at

Relationships:

- One-to-many: User (organizer) → Events, Event → Ticket Types, Order → Tickets
- Many-to-many: Events ↔ Categories (if multi-category support needed)
- Foreign keys with cascading rules (e.g., delete event → cascade delete tickets)

API Endpoints (Sample)

Authentication:

- POST /api/auth/register - User registration
- POST /api/auth/login - User login
- POST /api/auth/logout - User logout
- POST /api/auth/verify-email - Email verification
- POST /api/auth/reset-password - Password reset request
- POST /api/auth/reset-password/confirm - Complete password reset

Events:

- GET /api/events - List all events (with filters, search, pagination)
- GET /api/events/:id - Get event details
- POST /api/events - Create event (organizer only)
- PUT /api/events/:id - Update event (organizer only)
- DELETE /api/events/:id - Delete event (organizer only)
- GET /api/events/:id/analytics - Event analytics (organizer only)

Tickets:

- POST /api/orders - Create order (purchase tickets)
- GET /api/orders/:id - Get order details
- POST /api/orders/:id/checkout - Process payment
- GET /api/tickets/my-tickets - User's purchased tickets
- POST /api/tickets/:id/check-in - Check-in ticket (organizer only)

Venues:

- GET /api/venues - List all venues (with filters)
- GET /api/venues/:id - Get venue details

- POST /api/venues - Create venue (venue owner only)
- PUT /api/venues/:id - Update venue (venue owner only)
- POST /api/venues/:id/inquire - Send inquiry to venue owner

Admin:

- GET /api/admin/dashboard - Admin dashboard stats
- GET /api/admin/events/pending - Pending events for approval
- PUT /api/admin/events/:id/approve - Approve event
- PUT /api/admin/events/:id/reject - Reject event
- GET /api/admin/users - List all users
- GET /api/admin/transactions - List all transactions

Integration Points

Paymob Integration:

- API Endpoints: Authentication, Payment Key Generation, Transaction Inquiry
- Webhook: Receive payment status callbacks
- Testing: Sandbox environment with test cards

Fawry Integration:

- API Endpoints: Reference Number Generation, Payment Status Check
- Webhook: Payment confirmation callback
- Testing: Fawry test environment

Cloudinary Integration:

- Upload API: Image uploads with automatic optimization
- Transformation: Resize, crop, format conversion on-the-fly
- CDN: Global content delivery

SendGrid/Mailgun Integration:

- API: Send transactional emails
- Templates: Pre-designed email templates
- Webhooks: Track email opens, clicks, bounces

SMS Gateway Integration:

- API: Send SMS messages
 - Delivery reports: Track message delivery status
-

DEVELOPMENT TIMELINE

Month 1: Foundation & Core Infrastructure

Weeks 1-2:

- Requirements finalization and stakeholder approval (this document)
- Wireframes and UI mockups (Figma) - key pages: Homepage, Event Detail, Checkout, Dashboards
- Database schema design and entity-relationship diagram
- Technical architecture documentation
- Development environment setup (version control, CI/CD pipeline)
- Team onboarding (if outsourced development partner selected)

Weeks 3-4:

- Database setup (PostgreSQL + Redis)
- Backend API skeleton (Express.js, routing structure, middleware)
- Frontend setup (Next.js, Tailwind CSS, project structure)
- User authentication system (registration, login, JWT implementation)
- Basic user profile pages (Attendee, Organizer, Venue Owner dashboards)

Deliverable: Working authentication system (users can register, login, access role-specific dashboards)

Month 2: Ticketing Engine Core

Weeks 1-2:

- Event creation flow (organizer side) - Step 1: Basic info, Step 2: Tickets, Step 3: Review
- Event listing page (attendee side) - grid view with filters

- Event detail page - rich media, ticket types, purchase button
- Database integration (events, ticket_types tables)
- Image upload functionality (Cloudinary integration)

Weeks 3-4:

- Ticket purchase flow (cart → checkout → attendee info)
- Paymob payment gateway integration (sandbox environment)
- QR code generation (library integration, test QR codes)
- Order confirmation page
- Email notification system setup (SendGrid/Mailgun)
- Ticket purchase confirmation email with QR code

Deliverable: End-to-end ticket purchase working (create event → buy ticket → receive confirmation email with QR code) using sandbox payment

Month 3: Discovery, Search & Bilingual

Weeks 1-2:

- Event search functionality (full-text search in PostgreSQL)
- Filter system (date, location, category) - frontend UI and backend API
- Sorting options (date, price, popularity)
- Pagination (load more events)
- Related events algorithm (same category or venue)

Weeks 3-4:

- Internationalization (i18n) setup (next-i18next)
- Arabic translations for all UI elements
- RTL layout implementation (CSS direction switching, testing)
- Language toggle functionality
- Bilingual event data support (organizers can input Arabic + English)
- Paymob LIVE integration (production credentials, real payments)
- Live payment testing with small test transactions

Deliverable: Fully functional bilingual event discovery platform with real payment processing

Month 4: Venue Marketplace & Secondary Payment

Weeks 1-2:

- Venue listing creation form (venue owner side)
- Venue browsing page (organizer and attendee side)
- Venue detail page (photos, amenities, standardized pricing display)
- Venue search and filter functionality (capacity, location, amenities)
- Database integration (venues, venue_inquiries tables)

Weeks 3-4:

- Venue inquiry form (organizers send inquiries to venue owners)
- Venue owner dashboard (inquiry management interface)
- Email notifications for inquiries (venue owners notified immediately)
- Fawry payment gateway integration
- Fawry reference number generation and payment confirmation flow
- Premium venue listing feature (500 EGP/month subscription)

Deliverable: Venue marketplace live with inquiry system, Fawry payment option available

Month 5: Dashboards, Check-In & Analytics

Weeks 1-2:

- Organizer dashboard enhancements:
 - Real-time sales metrics
 - Visual charts (Chart.js integration)
 - Attendee list with export to CSV
 - Promo code creation and tracking
- Payout tracking interface (earnings breakdown, payout status)
- Admin panel development:

- Dashboard overview with platform-wide metrics
- Event approval queue
- Venue approval queue
- User management

Weeks 3-4:

- QR code check-in scanner (web-based, camera access)
- Check-in validation logic (prevent duplicate scans)
- Check-in feedback (visual and audio cues)
- SMS notification system integration (Vodafone SMS API or Twilio)
- SMS ticket confirmations and event reminders
- Email reminder system (24 hours before event)

Deliverable: Full organizer and admin tools operational, QR check-in system working

Month 6: Testing, Polish & Launch Prep

Weeks 1-2:

- Comprehensive testing:
 - Functional testing (all user flows)
 - Cross-browser testing (Chrome, Firefox, Safari, Edge, UC Browser)
 - Mobile device testing (iOS, Android, various screen sizes)
 - Payment gateway testing (Paymob and Fawry, success and failure scenarios)
 - Load testing (simulate 500 concurrent users)
- Bug fixing and issue resolution
- Security audit:
 - Penetration testing (third-party security firm if budget allows)
 - Vulnerability scanning (npm audit, OWASP checks)
 - SSL/TLS configuration review

Weeks 3-4:

- Performance optimization:

- Image optimization (compress, convert to WebP)
- Code minification and bundling
- Database query optimization (add indexes)
- CDN configuration (Cloudflare caching rules)
- Lighthouse audit (target score >85)
- SEO setup:
 - Meta tags for all pages
 - Sitemap generation and submission
 - Google Search Console setup
 - Structured data ([Schema.org](https://schema.org) JSON-LD)
- Beta testing with real users:
 - Recruit 10 event organizers (from your IT industry connections)
 - Onboard 100 venues (manual outreach, free listings)
 - Goal: 20-30 real events listed before public launch
- Marketing materials preparation:
 - Landing page copy and graphics
 - Social media accounts setup (Facebook, Instagram, LinkedIn)
 - Promotional video (30-60 seconds explainer)
 - Press release draft (for tech blogs)

Deliverable: Platform ready for public launch with 20-30 events listed, 100 venues onboarded

Launch Day & Post-Launch (End of Month 6)

Launch Activities:

- Go-live announcement on social media
- Email announcement to beta users and waitlist
- Facebook and Instagram ad campaigns start
- Press release distribution to Egyptian tech media
- Monitoring dashboard setup (real-time traffic and error tracking)

Post-Launch Support (First 2 Weeks):

- 24/7 monitoring for critical issues
- Daily bug triage and hotfix deployments

- User feedback collection (in-app surveys, support tickets)
 - Performance monitoring (identify bottlenecks)
 - Marketing campaign optimization (adjust targeting based on early results)
-

RISKS & MITIGATION

Technical Risks

Risk 1: Payment Gateway Integration Failures

- **Probability:** Medium
- **Impact:** High (cannot process payments = no revenue)
- **Mitigation:**
 - Dual payment gateway integration (Paymob + Fawry for redundancy)
 - Extensive testing in sandbox before going live
 - Fallback: Manual payment processing via bank transfer if both gateways fail temporarily
 - Monitoring: Real-time alerts for payment failures

Risk 2: Scalability Issues at Launch

- **Probability:** Medium
- **Impact:** Medium (slow performance frustrates users)
- **Mitigation:**
 - Load testing before launch (simulate 500-1000 concurrent users)
 - Horizontal scaling architecture (can add servers quickly)
 - CDN for static assets (reduces server load)
 - Database optimization (indexes, query optimization)
 - Auto-scaling: Use cloud infrastructure that scales automatically (Railway, DigitalOcean Kubernetes)

Risk 3: Security Breach or Data Leak

- **Probability:** Low (with proper security measures)
- **Impact:** Critical (reputation damage, legal liability)
- **Mitigation:**

- Security best practices from day one (HTTPS, encryption, secure authentication)
- Third-party security audit before launch
- Regular dependency updates (patch vulnerabilities)
- Incident response plan (documented steps for breach containment)
- Cyber insurance (consider for Phase 2)

Business Risks

Risk 4: Low Organizer Adoption (Chicken-and-Egg Problem)

- **Probability:** High (common marketplace challenge)
- **Impact:** High (no events = no attendees)
- **Mitigation:**
 - Pre-launch organizer outreach (target: 20 organizers committed before public launch)
 - First 10 organizers get ZERO commission (free trial period)
 - Leverage your IT industry connections for initial events (tech meetups, conferences)
 - Manual curation: Focus on quality over quantity initially
 - Success stories: Showcase early adopters in marketing

Risk 5: Competitive Response (Platinumlist Enters Egypt)

- **Probability:** Medium (they may notice and respond)
- **Impact:** High (well-funded competitor)
- **Mitigation:**
 - Speed to market: Launch quickly before they enter
 - Price moat: 3-5% commission difficult for them to match (hurts global margins)
 - Local advantage: Better Egypt localization, payment methods, partnerships
 - Venue marketplace: Unique feature they don't have (hard to replicate quickly)
 - Customer loyalty: Build strong relationships with early organizers

Risk 6: Regulatory Changes or Licensing Requirements

- **Probability:** Low-Medium (Egypt regulatory environment can change)
- **Impact:** Medium-High (could require business model changes)
- **Mitigation:**
 - Legal consultation: Engage Egyptian lawyer specializing in e-commerce/fintech
 - Compliance monitoring: Stay informed of regulatory updates
 - Industry associations: Join or monitor Egyptian tech and events industry groups
 - Terms of service: Clear disclaimer that organizers responsible for event permits and licenses
 - Escrow model: Platform holds funds, not acting as payment processor (reduces financial regulation exposure)

Operational Risks

Risk 7: Outsourced Development Partner Underperforms

- **Probability:** Medium (common outsourcing challenge)
- **Impact:** High (delays, quality issues)
- **Mitigation:**
 - Thorough vetting: Check references, review portfolio, assess technical skills
 - Milestone-based payments: Pay only upon deliverable acceptance
 - Weekly demos: Require working demos every Friday (catch issues early)
 - Code reviews: Technical advisor reviews code quality periodically
 - Contract clauses: Include performance guarantees and termination rights
 - Backup plan: Identify alternative agencies in advance

Risk 8: Key Person Dependency (You)

- **Probability:** Medium (if you're the sole decision-maker)
- **Impact:** Medium (bottlenecks, delays)
- **Mitigation:**

- Document everything: Clear requirements, processes, decisions
 - Delegate: Empower development partner or co-founder to make tactical decisions
 - Communication protocols: Clear escalation paths for urgent issues
 - Succession planning: Identify potential co-founder or technical lead early
-

SUCCESS CRITERIA

MVP Success Definition

The MVP will be considered successful if, by the end of Month 6:

1. Platform Functionality:

- All P0 (Must Have) features operational and tested
- Zero critical bugs in production
- Payment processing working reliably (>99% success rate)
- Platform accessible 99%+ uptime

2. User Acquisition:

- 100 venues listed (Cairo and Alexandria)
- 30 active organizers (posted at least one event)
- 50 published events (mix of upcoming and recurring)
- 1,500 registered attendees
- 2,000 tickets sold (cumulative from launch)

3. Financial Metrics:

- 300,000 EGP GMV (Gross Merchandise Value - total ticket sales)
- 15,000 EGP platform revenue (5% average commission)
- 10,000 EGP from venue premium listings (20 venues × 500 EGP/month)
- Positive contribution margin (revenue > variable costs)

4. User Satisfaction:

- Net Promoter Score (NPS) >40 (measured via post-purchase survey)
- <5% refund rate (low refunds = satisfied attendees)
- Average 4+ star rating from organizers (platform usability survey)

- <2% churn rate for organizers (repeat event creation)

5. **Operational Excellence:**

- Average event approval time <24 hours
- Customer support response time <12 hours (during business hours)
- Payment dispute rate <1% of transactions
- Check-in success rate >95% (QR codes scan correctly)

Phase 2 Readiness Indicators

The MVP will be ready for Phase 2 expansion (GCC entry, advanced features) when:

- Proven product-market fit in Egypt (consistent month-over-month growth in ticket sales)
- Stable platform performance (no major technical debt blocking scale)
- Established brand reputation (positive media coverage, word-of-mouth)
- Operational processes documented (onboarding, support, content moderation can be replicated)
- Financial sustainability (revenue covers fixed costs, path to profitability clear)

APPENDICES

Appendix A: User Flow Diagrams

(To be created in Figma or similar tool during Month 1)

Key Flows:

1. Attendee: Registration → Event Discovery → Ticket Purchase → Check-In
2. Organizer: Registration → Event Creation → Sales Monitoring → Check-In Management → Payout
3. Venue Owner: Registration → Venue Listing → Inquiry Management → Premium Upgrade
4. Admin: Login → Event Approval → User Management → Transaction Monitoring

Appendix B: Wireframes & Mockups

(To be created during Month 1)

Priority Pages for Wireframes:

1. Homepage (Desktop + Mobile)
2. Event Listing Page
3. Event Detail Page
4. Checkout Flow (3 steps)
5. Organizer Dashboard
6. Venue Detail Page
7. Admin Panel

Appendix C: API Documentation

(To be created during development, published using Swagger or Postman)

Documentation Requirements:

- All API endpoints documented with request/response schemas
- Authentication requirements specified
- Error codes and messages listed
- Code examples in JavaScript (frontend consumption)

Appendix D: Brand Guidelines

Platform Name: *[To be decided - e.g., "Tazkara", "EventHub Egypt", "Hafla"]*

Brand Colors:

- Primary: [Color HEX code]
- Secondary: [Color HEX code]
- Accent: [Color HEX code]

Typography:

- Arabic: Cairo, Tajawal, or Almarai (Google Fonts)
- English: Inter, Open Sans, or Roboto

Logo: [To be designed]

Tone of Voice:

- Friendly and approachable
- Clear and concise (avoid jargon)
- Bilingual (equal emphasis on Arabic and English)
- Professional for organizer/venue owner communications
- Enthusiastic for attendee-facing content

Appendix E: Glossary

Key Terms:

- **Attendee:** End-user who purchases tickets to attend events
 - **Organizer:** User who creates and manages events, sells tickets
 - **Venue Owner:** User who lists venue properties for rental
 - **GMV (Gross Merchandise Value):** Total value of ticket sales (before commission deduction)
 - **Commission:** Platform's revenue share from ticket sales (3-5%)
 - **Payout:** Transfer of event revenue (minus commission) to organizer's bank account
 - **Inquiry:** Booking request sent from organizer to venue owner
 - **Premium Listing:** Paid feature for venues (featured placement)
 - **Promo Code:** Discount code created by organizers for ticket promotions
 - **QR Code:** Quick Response code on tickets for check-in scanning
 - **Check-In:** Process of validating and admitting attendees at event entry
 - **MVP (Minimum Viable Product):** Initial product version with core features only
 - **P0/P1/P2:** Priority levels (P0 = Must Have, P1 = Should Have, P2 = Nice to Have)
 - **PWA (Progressive Web App):** Web application that functions like native mobile app
 - **RTL (Right-to-Left):** Layout direction for Arabic language
 - **i18n (Internationalization):** Technical term for multi-language support
-

DOCUMENT APPROVAL

This Product Requirements Document is submitted for review and approval by key stakeholders before development commences.

Prepared By: Product Team

Date: February 13, 2026

Approval Required From:

Business Owner / Founder

Technical Lead / CTO (or outsourced development partner)

Legal Advisor (Egypt e-commerce compliance review)

Financial Controller (budget and revenue model approval)

Revision History:

Version	Date	Changes
1.0	Feb 13, 2026	Initial MVP document

END OF DOCUMENT

This is a living document. Updates and refinements will be tracked through version control as development progresses and requirements evolve based on user feedback and market conditions.